Controller Design for Flapping Flight

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We propose a control strategy for flapping flight.

2 Model Description

The lift and drag coefficients are defined as

$$C_L = C_{L,0} \sin 2\alpha \tag{1}$$
$$C_D = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \tag{2}$$

We denote the nondimensional phase as $\tau^* = \omega^* t^*$. A wingbeat cycle occurs between $\tau^* = 0$ and $\tau^* = 2\pi$. The flapping kinematics are

$$s^*(\tau^*) = s_0^* \sin \tau^* \tag{3}$$

The pitch kinematics are

$$\psi(\tau^*) = \psi_0 + \psi_1 \sin(\tau^* - \delta_0) \tag{4}$$

3 Controller Design

3.1 Problem Statement

We wish to design a control strategy to achieve arbitrary trajectories in the two-dimensional plane, including prey pursuit trajectories. We assume that the flyer's own state is known using a variety of sensory inputs, so position and velocity are available as inputs to the controller. In addition, when a prey is present in the environment, the bearing angle to the prey, as detected by the eyes, is also an available input. Because we are particularly interested in *dragonfly* flight, we make two design decisions such that the resulting flight characteristics resemble those of a dragonfly. First, hovering should be achieved with an inclined stroke plane (tilted from the horizontal), characterized by the parameter γ^{hover}. Second, when at speed, the stroke-plane normal should be approximately aligned with the direction of travel, as observed in experiments [1]. This last choice will simplify the analysis to follow, since any flow velocity component parallel to the stroke plane can be neglected. With these choices in mind, the contents of this section otherwise hold for flapping flight in general, as we do not assume specific morphological characteristics like number of wing pairs, size, etc.

3.2 Mean Aerodynamic Force

3.2.1 Definitions and Assumptions

To affect its trajectory, the flapping flyer must modulate and direct the aerodynamic force acting on its wings. Because the aerodynamic force on a flapping wing is by nature highly unsteady, it is convenient to consider the wingbeat-cycle average of the aerodynamic force, summed over each of the wings, which we call the *mean aerodynamic force*, $\langle \vec{F}^* \rangle$. $\langle \vec{F}^* \rangle$ is a function of morphology and wing kinematics, as well as the flow characteristics in the body frame (this involves the body kinematics in the fixed frame, as well as any wind in the fixed frame; in the following we refer to this simply as the body kinematics, or body velocity, with the implication that it is relative to the air, which need not be the same as the fixed frame if there is any wind). Body velocity is not constant over a wingbeat cycle: there are (a) oscillations inherent to flapping, and there may be (b) a net change in body velocity over the cycle. Assuming the oscillations are small compared to the characteristic wing velocity, and that any body velocity changes occur at a slow rate relative to the wingbeat period, the body velocity may be considered constant over a wingbeat cycle. The following assumes these conditions are met. Our objective is now to understand how $\langle \vec{F}^* \rangle$ depends on the wing kinematics at any given body velocity, so that we are able to formulate a control strategy.

As explained in subsection 3.1, we are designing a controller under the assumption that the body velocity is normal to the stroke plane. Beyond morphology and kinematics, then, there is a single body velocity parameter $\langle u^* \rangle$, which is the cycle-averaged body velocity magnitude (the direction being set to the stroke-plane normal $\hat{n}$). It is convenient to introduce a similarity parameter in the form of the *advance ratio*, J, which is the ratio of the cycle-averaged body velocity to the characteristic wing velocity:

$$J = \frac{\langle u^* \rangle}{s_0^* \omega^*} \tag{5}$$

The instantaneous wing velocity $\vec{v}^*$ relative to the air has the stroke-plane normal and in-plane components

$$v_n^* = \langle u^* \rangle = s_0^* \omega^* J \tag{6}$$
$$v_s^* = s_0^* \omega^* \cos \tau^* \tag{7}$$

The squared norm of the velocity is

$$(v^*)^2 = \left(s_0^* \omega^*\right)^2 \left[J^2 + \cos^2 \tau^*\right] \tag{8}$$

We introduce the signed angle χ from $\hat{n}$ to $\vec{v}^*$, defined as

$$\chi = -\tan^{-1} \frac{v_s^*}{v_n^*} = -\tan^{-1} \frac{\cos \tau^*}{J} \tag{9}$$

When $\cos \tau^* > 0$ (stroke along $+\hat{s}$, $\chi < 0$), the angle of attack α is

$$\alpha = \psi - \chi = \psi + \tan^{-1} \frac{\cos \tau^*}{J} \tag{10}$$

When $\cos \tau^* < 0$ (stroke along $-\hat{s}$, $\chi > 0$), the angle of attack α is

$$\alpha = \chi - \psi = -\psi - \tan^{-1} \frac{\cos \tau^*}{J} \tag{11}$$

The general expression is, then,

$$\alpha = \frac{\cos \tau^*}{|\cos \tau^*|} \psi + \tan^{-1} \frac{|\cos \tau^*|}{J} \tag{12}$$

The above is sufficient to obtain the mean aerodynamic force. The derivation is somewhat involved in the general case. We first present the hovering case, which corresponds to $J = 0$.

3.2.2 Hover Case ($J = 0$)

For $J = 0$, the angle of attack simplifies to

$$\alpha = \frac{\cos \tau^*}{|\cos \tau^*|} \psi + \frac{\pi}{2} \tag{13}$$

Using the appropriate trigonometric identities, the aerodynamic coefficients (defined in Equation 1 and Equation 2) are

$$C_L = -\frac{\cos\tau^*}{|\cos\tau^*|} C_{L,0} \sin 2\psi \tag{14}$$

$$C_D = \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \tag{15}$$

Because the wing velocity relative to the air is purely in-plane (along $\hat{s}$), the in-plane aerodynamic force component is purely drag, and the out-of-plane component (along $\hat{n}$) is purely lift. The mean aerodynamic force components are

$$\begin{aligned}
\langle F_n^* \rangle &= \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_L |\cos\tau^*|^2 \rangle \\
&= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 C_{L,0} \langle \sin 2\psi \ \cos\tau^* |\cos\tau^*| \rangle \qquad (16) \\
\langle F_s^* \rangle &= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_D \cos\tau^* |\cos\tau^*| \rangle \\
&= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \ \cos\tau^* |\cos\tau^*| \rangle \qquad (17)
\end{aligned}$$

Note the constant drag term vanishes because $\langle \cos\tau^* |\cos\tau^*| \rangle = 0$. We introduce the shorthand $\Delta C_D = C_{D,\frac{\pi}{2}} - C_{D,0}$. The expressions feature a common pre-factor we call q^*, defined as

$$q^* = \frac{1}{4}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \tag{18}$$

We further introduce the normal and in-plane force coefficients $C_{F_n^*}$ and $C_{F_s^*}$, defined as

$$C_{F_n^*} = \frac{F_n^*}{q^*} = -2C_{L,0} \langle \sin 2\psi \ \cos\tau^* |\cos\tau^*| \rangle \tag{19}$$

$$C_{F_s^*} = \frac{F_s^*}{q^*} = -\Delta C_D \langle \cos 2\psi \ \cos\tau^* |\cos\tau^*| \rangle \tag{20}$$

which we synthesize into the overall force magnitude coefficient

$$C_{F^*} = \sqrt{C_{F_n^*}^2 + C_{F_s^*}^2} \tag{21}$$

This coefficient is useful because it is only a function of the pitch kinematic parameters and the lift and drag coefficient parameters, and is independent of morphology and wingbeat frequency and amplitude, which appear only in the pre-factor q^*. By maximizing the value of this coefficient, given a constraint on the force direction required to achieve a given maneuver, the characteristic wingbeat velocity $s_0^*\omega^*$ can be minimized. With our parametrization of the pitch kinematics, and with the lift and drag coefficient parameters set, C_{F^*} and its (n, s) decomposition are functions of 3 variables: ψ_0, ψ_1, and δ_0. We visualize C_{F^*}, $C_{F_n^*}$, and $C_{F_s^*}$ across slices of this 3D space in Figure 1 and Figure 2.

As panel (a) of Figure 2 makes evident, significantly more force is produced when $\delta_0 = \pm 90°$. Because $\delta_0 = -90°$ results in the mean force pointing along $-\hat{n}$ when the mean pitch is neutral ($\psi_0 = 0°$), we select $\delta_0 = +90°$ as the operating point. This choice maximizes the mean force for a given (ψ_0, ψ_1), and reduces the pitch parameter space to 2D. How then do we select ψ_0 and ψ_1? It is clear from panel (b) that a certain value of ψ_1, around $\psi_1 \approx 50°$, maximizes C_{F^*} for any given ψ_0. We will call it ψ_1^{opt}. In addition, $\psi_0 = 0°$ (mod 90°) maximizes C_{F^*}. Looking at panel (b) of Figure 1, it is clear that the in-plane force component is

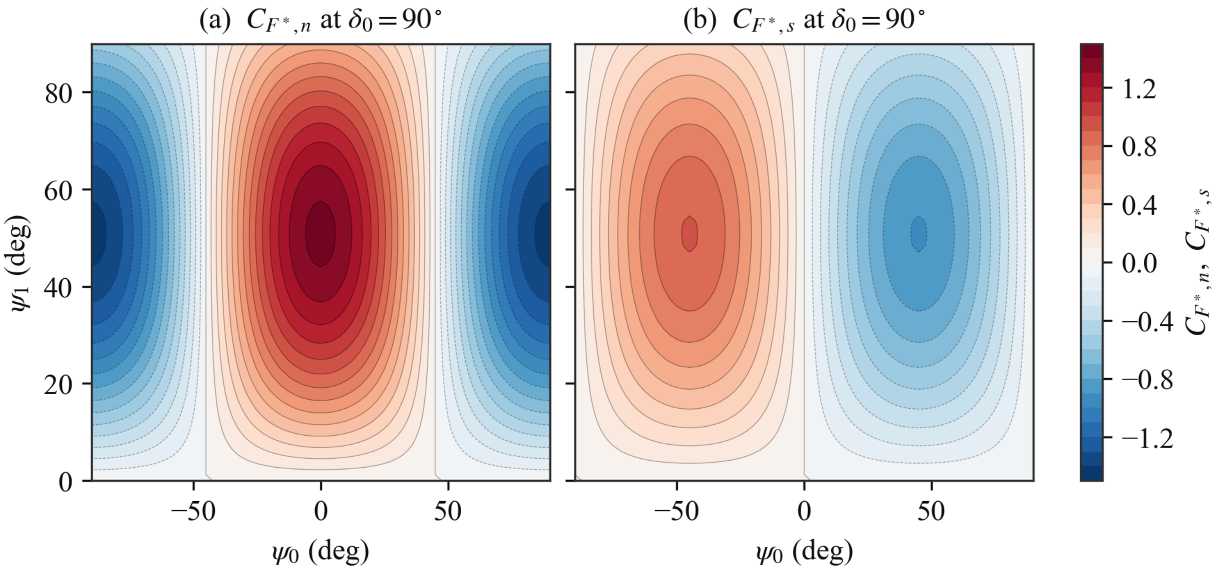

Figure 1: (a) Stroke-plane-normal (lift) component $C_{F_n^*}$ and (b) in-plane (drag) component $C_{F_s^*}$ of the force coefficient over the pitch plane at $\delta_0 = 90^\circ$. The force direction quadrant in the $(\hat{s}, \hat{n})$ plane can be inferred from the signs.

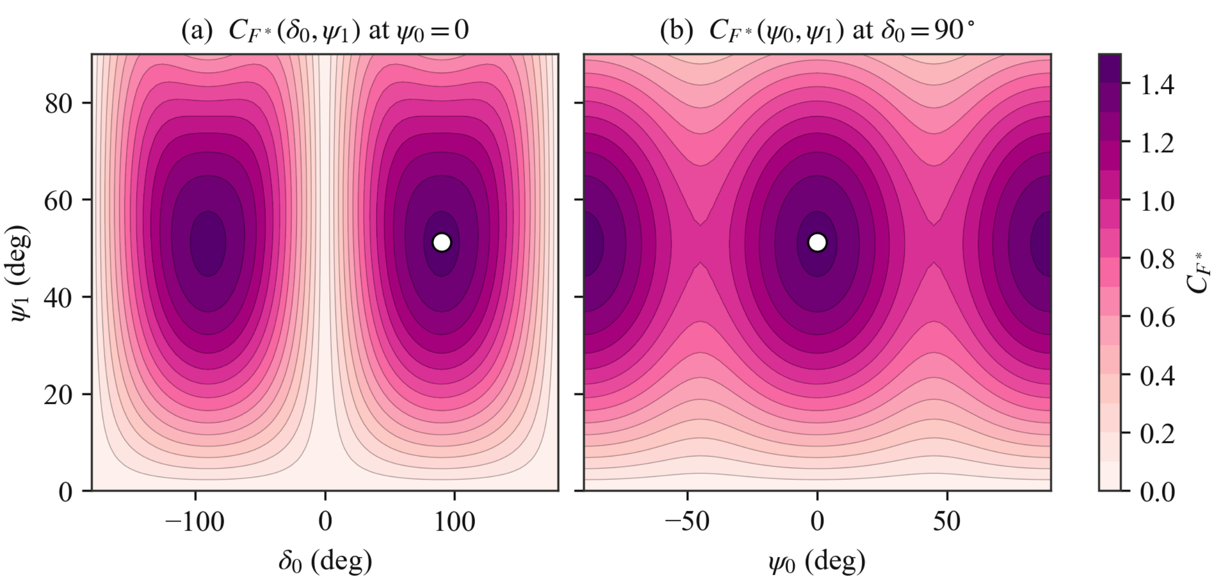

Figure 2: Dependence of C_{F^*} on pitch parameters at zero body velocity. Maximum shown as white dots.

zero when $\psi_0 = 0° \pmod{90°}$, so the force is purely normal to the stroke plane. In addition, from panel (a) of the same figure, we see that $\psi_0 = 0°$ results in the mean force pointing towards $+\hat{n}$, while $\psi_0 = \pm 90°$ results in the mean force pointing towards $-\hat{n}$. We select neutral pitch ($\psi_0 = 0°$) as the reference, for consistency with the choice of $\delta_0 = 90°$. In summary, the reference pitch kinematics

$$\psi_0 = 0° \qquad \psi_1 = \psi_1^{\mathrm{opt}} \qquad \delta_0 = 90° \tag{22}$$

result in $C_{F^*} = C_{F^*}^{\max}$, and $\langle \vec{F}^* \rangle$ pointing towards $+\hat{n}$. These reference kinematics allow for *efficient* hover (in the sense that the characteristic wingbeat velocity $s_0^* \omega^*$ is minimized, because C_{F^*} is maximized) when the stroke plane is horizontal ($\gamma^{\mathrm{hover}} = 0°$). Indeed, since the mean force vector points towards $+\hat{n}$, $+\hat{n}$ must be vertical and opposed to the direction of gravity. To model dragonfly hover specifically, at an inclined stroke plane ($\gamma^{\mathrm{hover}} \neq 0°$), the mean force vector must have an in-plane component. Looking at panel (b) of Figure 1, it is clear that this can be achieved by changing ψ_0, away from the symmetric 0° point. For convenience, we introduce β, the signed angle from the stroke-plane normal $\hat{n}$ to the mean force vector. Since the mean force is $\langle \vec{F}^* \rangle = q^* \left(C_{F_n^*} \hat{n} + C_{F_s^*} \hat{s} \right)$, β can be expressed from the $C_{F_n^*}$ and $C_{F_s^*}$ coefficients (Equation 19 and Equation 20) as

$$\beta = -\tan^{-1} \frac{C_{F_s^*}}{C_{F_n^*}} = -\tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \frac{\langle \cos 2\psi \cos \tau^* |\cos \tau^*| \rangle}{\langle \sin 2\psi \cos \tau^* |\cos \tau^*| \rangle} \right] \tag{23}$$

We can simplify the $\langle \cos 2\psi \cos \tau^* |\cos \tau^*| \rangle / \langle \sin 2\psi \cos \tau^* |\cos \tau^*| \rangle$ ratio in the above expression. With $\delta_0 = 90°$, the pitch kinematics reduce to $\psi = \psi_0 - \psi_1 \cos \tau^*$. Using the appropriate trigonometric identities,

$$\cos 2\psi = \cos 2\psi_0 \cos(2\psi_1 \cos \tau^*) + \sin 2\psi_0 \sin(2\psi_1 \cos \tau^*) \tag{24}$$

$$\sin 2\psi = \sin 2\psi_0 \cos(2\psi_1 \cos \tau^*) - \cos 2\psi_0 \sin(2\psi_1 \cos \tau^*) \tag{25}$$

Under the half-period shift $\tau^* \to \tau^* + \pi$, the factors $\cos \tau^* |\cos \tau^*|$ and $\sin(2\psi_1 \cos \tau^*)$ change sign while $\cos(2\psi_1 \cos \tau^*)$ is unchanged, so the $\cos(2\psi_1 \cos \tau^*)$ terms average to zero, leading to

$$\langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle = \sin 2\psi_0 \langle \sin(2\psi_1 \cos \tau^*) \cos \tau^* |\cos \tau^*| \rangle \tag{26}$$

$$\langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle = -\cos 2\psi_0 \langle \sin(2\psi_1 \cos \tau^*) \cos \tau^* |\cos \tau^*| \rangle \tag{27}$$

The common $\langle \sin(2\psi_1 \cos \tau^*) \cos \tau^* |\cos \tau^*| \rangle$ factor cancels out in the ratio, giving simply

$$\frac{\langle \cos 2\psi \ \cos \tau^* |\cos \tau^*| \rangle}{\langle \sin 2\psi \ \cos \tau^* |\cos \tau^*| \rangle} = -\tan 2\psi_0 \tag{28}$$

and Equation 23 becomes

$$\beta = \tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \tan 2\psi_0 \right] \tag{29}$$

Importantly, β is independent of ψ_1, and is only controlled by ψ_0. It follows that ψ_0 is an appropriate control variable for the mean force direction relative to the stroke-plane normal $\hat{n}$. β is shown as a function of ψ_0 in Figure 3. As discussed above, C_{F^*} becomes less than its maximum when $\psi_0 \neq 0° \pmod{90°}$, away from the reference pitch kinematics, so it is also shown in addition to β. In the $\Delta C_D / (2C_{L,0}) \to 1$ limit, the relation between β and ψ_0 can be approximated as $\beta \approx 2\psi_0$. This holds only roughly here because $\Delta C_D / (2C_{L,0}) = 1.9/3 \approx 0.63$, but the trend is the same, as can be seen in the figure. To achieve hover at an inclined stroke plane with tilt angle γ^{hover}, the mean force vector angle must be $\beta = -\gamma^{\mathrm{hover}}$. This requires $\psi_0 = \psi_0^{\mathrm{hover}}$, given by Equation 29 (specifically the inverse relation) when $\beta = -\gamma^{\mathrm{hover}}$.

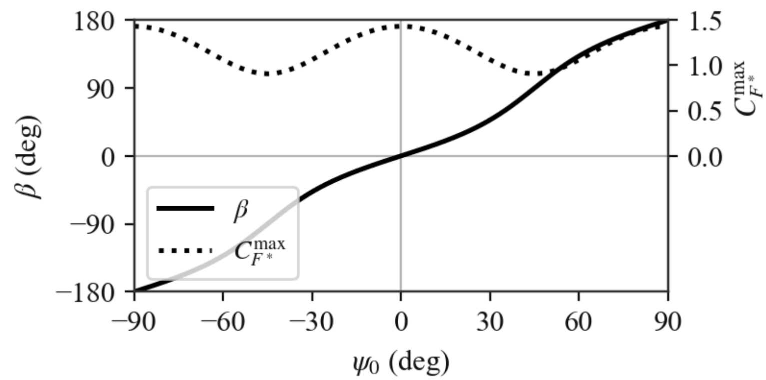

Figure 3: Effect of ψ_0 on force direction angle β and force coefficient C_{F^*}.

Now that we have narrowed down the pitch kinematics for hover, we step back to consider the flapping kinematics as well, governed by parameters s_0^* (amplitude) and ω^* (frequency). The condition for hover on the mean force magnitude is

$$C_{F^*} q^* = 1 \implies (s_0^* \omega^*)^2 = \frac{4}{C_{F^*} \frac{A^*}{m^*}} \tag{30}$$

We prefer to consider ω^* as a fixed parameter rather than a control variable, leaving s_0^* as the primary control for the force amplitude. The hover value is given by

$$(s_0^*)^{\text{hover}} = \frac{1}{\omega^*} \sqrt{\frac{4}{C_{F^*}^{\text{hover}} \frac{A^*}{m^*}}} \tag{31}$$

The hover kinematic parameters are summarized in Table 1. The control variables are s_0^* and ψ_0.

Table 1: Wing kinematic parameters for hover

Parameter	Value	Variables				
s_0^*	$(s_0^*)^{\text{hover}}$	$\frac{A^*}{m^*}$	ω^*	γ^{hover}	$C_{L,0}$	ΔC_D
ψ_0	ψ_0^{hover}			γ^{hover}	$C_{L,0}$	ΔC_D
ψ_1	ψ_1^{opt}					None
δ_0	90°					None

3.2.3 General Case ($J \geq 0$)

For $J \neq 0$, the wing velocity relative to the air is no longer purely in-plane, so lift and drag each contribute to both force components. In terms of the angle χ, the wing velocity direction is $\hat{v} = \cos\chi\,\hat{n} - \sin\chi\,\hat{s}$. Drag acts along $-\hat{v}$, and lift acts along the perpendicular direction

$$\hat{\ell} = -\sigma\,(\sin\chi\,\hat{n} + \cos\chi\,\hat{s}) \qquad \sigma \equiv \frac{\cos\tau^*}{|\cos\tau^*|} \tag{32}$$

whose sense alternates between half-strokes, consistent with the sign convention used to define α (in the hover case, $\hat{\ell}$ reduces to $+\hat{n}$ on both half-strokes). The instantaneous force is

$$\vec{F}^* = \frac{1}{2}\frac{A^*}{m^*}(v^*)^2 \left(C_L\,\hat{\ell} - C_D\,\hat{v}\right) \tag{33}$$

with components

$$F_n^* = \frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(-\sigma\sin\chi\; C_L - \cos\chi\; C_D) \tag{34}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(\sigma\cos\chi\; C_L - \sin\chi\; C_D) \tag{35}$$

Substituting $\cos\chi = J/\sqrt{J^2+\cos^2\tau^*}$, $\sin\chi = -\cos\tau^*/\sqrt{J^2+\cos^2\tau^*}$, and the expression for $(v^*)^2$ (Equation 8), we get

$$F_n^* = \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2+\cos^2\tau^*}\;(|\cos\tau^*|\; C_L - J\; C_D) \tag{36}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2+\cos^2\tau^*}\;(\sigma J\; C_L + \cos\tau^*\; C_D) \tag{37}$$

Writing $\theta \equiv \tan^{-1}(|\cos\tau^*|/J)$, such that $\alpha = \sigma\psi + \theta$ (see Equation 12),

$$\cos 2\theta = \frac{J^2-\cos^2\tau^*}{J^2+\cos^2\tau^*} \qquad \sin 2\theta = \frac{2J|\cos\tau^*|}{J^2+\cos^2\tau^*} \tag{38}$$

and the aerodynamic coefficients (Equation 1 and Equation 2) become

$$C_L = C_{L,0}\,(\sigma\cos 2\theta\;\sin 2\psi + \sin 2\theta\;\cos 2\psi) \tag{39}$$

$$C_D = \bar{C}_D - \frac{\Delta C_D}{2}\,(\cos 2\theta\;\cos 2\psi - \sigma\sin 2\theta\;\sin 2\psi) \tag{40}$$

where we define $\bar{C}_D \equiv \left(C_{D,\frac{\pi}{2}} + C_{D,0}\right)/2$ for brevity. Substituting into the force components, using $\sigma|\cos\tau^*| = \cos\tau^*$ and $\sigma^2 = 1$, and averaging, the force coefficients are

$$\begin{aligned} C_{F_n^*} &= 2\left\langle \frac{\left[C_{L,0}\left(J^2-\cos^2\tau^*\right) - \Delta C_D\, J^2\right]\cos\tau^*\;\sin 2\psi}{\sqrt{J^2+\cos^2\tau^*}}\right\rangle \\ &\quad + 2\left\langle \frac{J\left[2C_{L,0}\cos^2\tau^* + \frac{\Delta C_D}{2}\left(J^2-\cos^2\tau^*\right)\right]\cos 2\psi}{\sqrt{J^2+\cos^2\tau^*}}\right\rangle - 2J\bar{C}_D\left\langle\sqrt{J^2+\cos^2\tau^*}\right\rangle \end{aligned} \tag{41}$$

$$\begin{aligned} C_{F_s^*} &= -2\left\langle \frac{J\left[C_{L,0}\left(J^2-\cos^2\tau^*\right) + \Delta C_D\cos^2\tau^*\right]\sin 2\psi}{\sqrt{J^2+\cos^2\tau^*}}\right\rangle \\ &\quad - 2\left\langle \frac{\left[2C_{L,0}J^2 - \frac{\Delta C_D}{2}\left(J^2-\cos^2\tau^*\right)\right]\cos\tau^*\;\cos 2\psi}{\sqrt{J^2+\cos^2\tau^*}}\right\rangle \end{aligned} \tag{42}$$

Note the constant ($\bar{C}_D$) drag term of $C_{F_s^*}$ vanishes because $\langle\cos\tau^*\sqrt{J^2+\cos^2\tau^*}\rangle = 0$. Setting $J = 0$ recovers Equation 19 and Equation 20, as expected. At the operating point $\delta_0 = 90°$, $\psi = \psi_0 - \psi_1\cos\tau^*$, and the parity argument of the hover case applies unchanged: under $\tau^* \to \tau^* + \pi$, the bracketed polynomials and $\sqrt{J^2+\cos^2\tau^*}$ are invariant, so expanding $\sin 2\psi$ and $\cos 2\psi$ and discarding the vanishing averages leaves

$$C_{F_n^*} = 2\,I_n(J,\psi_1)\;\cos 2\psi_0 - 2J\bar{C}_D\,I_0(J) \tag{43}$$

$$C_{F_s^*} = -2\,I_s(J,\psi_1)\;\sin 2\psi_0 \tag{44}$$

with

$$I_0 = \left\langle \sqrt{J^2 + \cos^2 \tau^*} \right\rangle \tag{45}$$

$$\begin{aligned} I_n = &\left\langle \frac{J \left[2C_{L,0} \cos^2 \tau^* + \frac{\Delta C_D}{2} \left(J^2 - \cos^2 \tau^* \right) \right] \cos\left(2\psi_1 \cos \tau^*\right)}{\sqrt{J^2 + \cos^2 \tau^*}} \right\rangle \\ &- \left\langle \frac{\left[C_{L,0} \left(J^2 - \cos^2 \tau^* \right) - \Delta C_D \, J^2 \right] \cos \tau^* \, \sin\left(2\psi_1 \cos \tau^*\right)}{\sqrt{J^2 + \cos^2 \tau^*}} \right\rangle \end{aligned} \tag{46}$$

$$\begin{aligned} I_s = &\left\langle \frac{J \left[C_{L,0} \left(J^2 - \cos^2 \tau^* \right) + \Delta C_D \cos^2 \tau^* \right] \cos\left(2\psi_1 \cos \tau^*\right)}{\sqrt{J^2 + \cos^2 \tau^*}} \right\rangle \\ &+ \left\langle \frac{\left[2C_{L,0} J^2 - \frac{\Delta C_D}{2} \left(J^2 - \cos^2 \tau^* \right) \right] \cos \tau^* \, \sin\left(2\psi_1 \cos \tau^*\right)}{\sqrt{J^2 + \cos^2 \tau^*}} \right\rangle \end{aligned} \tag{47}$$

The angle from the stroke-plane normal to the mean force (Equation 23) becomes

$$\beta = \tan^{-1} \left[\frac{I_s \sin 2\psi_0}{I_n \cos 2\psi_0 - J \bar{C}_D \, I_0} \right] \tag{48}$$

In the limit $J \to 0$, the I_0 term vanishes and

$$I_n \to C_{L,0} \left\langle \sin\left(2\psi_1 \cos \tau^*\right) \cos \tau^* |\cos \tau^*| \right\rangle \qquad I_s \to \frac{\Delta C_D}{2} \left\langle \sin\left(2\psi_1 \cos \tau^*\right) \cos \tau^* |\cos \tau^*| \right\rangle \tag{49}$$

recovering Equation 29. The overall force coefficient C_{F^*} is shown in Figure 4 for values of J from 0 to 1, at $\delta_0 = 90°$. Highlighted on each panel is the reference point from the hover plot Figure 2, tracked as it turns from a global maximum at $J = 0$ to a local maximum, and eventually to a saddle point. This point represents the maximum force available in the $+\hat{n}$ direction for a given J. The region of space within the dotted contour around the reference point corresponds to $C_{F_n^*} > 0$. Within this region, the stroke-plane-normal component of the mean force points towards $+\hat{n}$.

The values of C_{F^*} and ψ_1 at the reference point are shown as functions of J in Figure 5. As the figure makes clear, to obtain the maximum force along $+\hat{n}$, ψ_1 should be decreased as J increases. The mechanism behind this is illustrated in Figure 6. For a given ψ, α decreases as J is increased. This moves us away from the ideal α, reducing lift, and in turn the force component along $+\hat{n}$. By decreasing ψ, the ideal α is recovered, thrust along $+\hat{n}$ is maximized. Of course, this is an instantaneous picture, but the conclusion can be extended to the mean force over the wingbeat stroke. To achieve the required reduction in ψ everywhere throughout the stroke, it is hopefully clear that the relevant parameter is ψ_1.

Finally, Figure 7 shows β as a function of ψ_0 for J from 0 to 1. Specifically, since β now depends on ψ_1 in addition to ψ_0 (see Equation 48), each line is obtained using the ψ_1 value adapted to the value of J (as shown in Figure 4). As J is increased, there is a clear steepening of the slope around $\psi_0 = 0$, meaning control authority is significantly increased. This would tend to suggest that any gain associated with the control variable ψ_0 may need to adapt based on the current J.

We conclude this analytical section by summarizing the kinematic parameters for the general case, in Table 2. Since there is no requirement for steady flight like in hover, no specific values for the control variables s_0^* and ψ_0 are prescribed, however the relevant variables that would inform the choice of the values are still listed.

3.3 Control Strategy

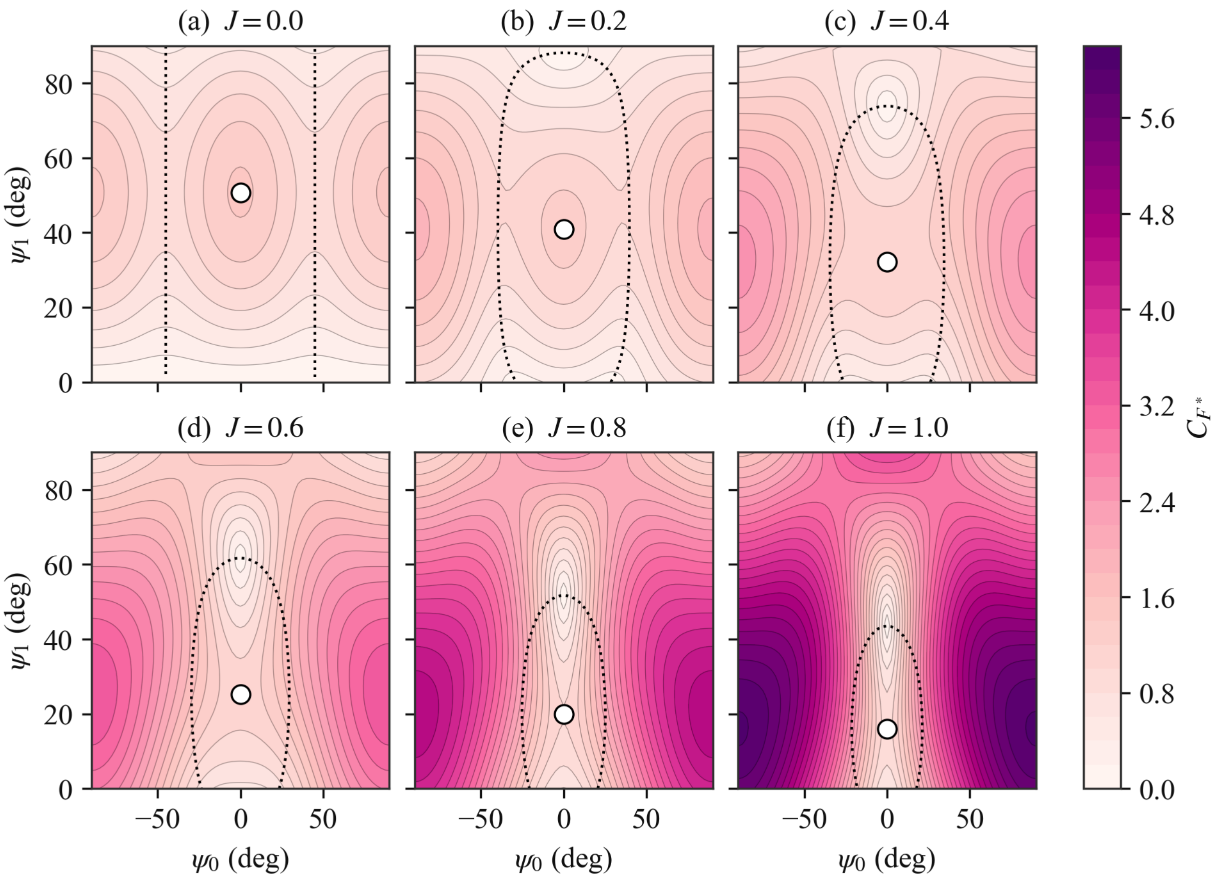

Figure 4: Force coefficient C_{F^*} over the pitch plane (ψ_0, ψ_1) at $\delta_0 = 90°$, for increasing advance ratio J. White dots track the hover optimum as it moves with J. It remains on the $\psi_0 = 0°$ axis by symmetry while ψ_1 decreases, and it turns from a local maximum into a saddle point. The global maximum sits at large $|\psi_0|$, where the force is dominated by drag from the normal inflow and points along $-\hat{n}$. The dotted contour marks $C_{F_n^*} = 0$, the boundary between propulsive ($+\hat{n}$) and braking ($-\hat{n}$) mean force.

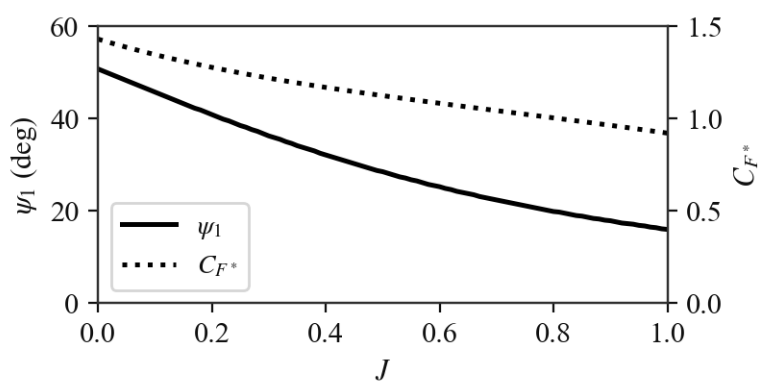

Figure 5: Pitch amplitude ψ_1 and force coefficient C_{F^*} of the continued hover optimum (the point tracked in Figure 4) as functions of advance ratio J. The point turns from a local maximum into a saddle at $J \approx 0.46$.

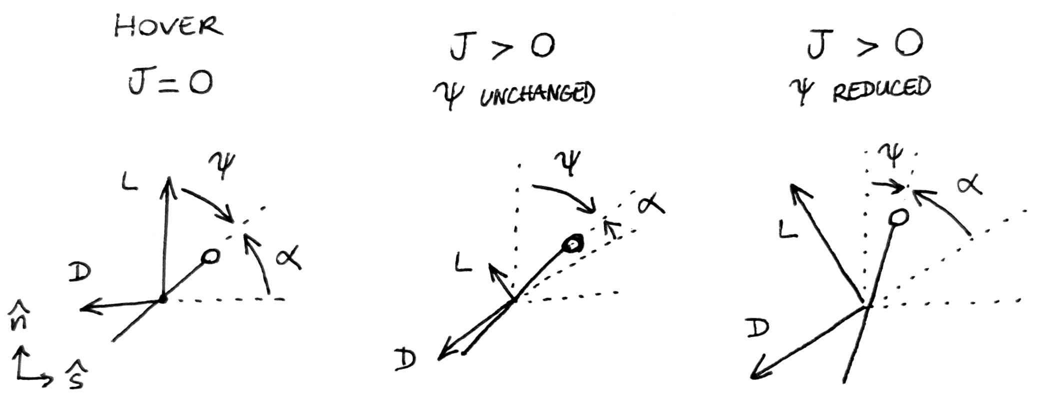

Figure 6: Illustration of the ψ reduction mechanism for maximizing stroke-plane-normal force as J increases.

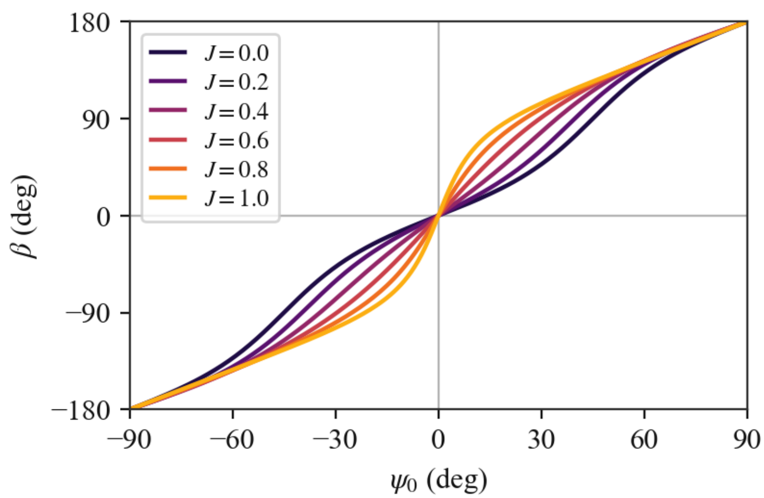

Figure 7: Effect of ψ_0 on the force direction angle β for increasing advance ratio J, at $\delta_0 = 90°$ and with ψ_1 set, for each J, to the continued-optimum value of Figure 5. Unlike in the hover case, β depends on ψ_1 when $J > 0$.

Table 2: Wing kinematic parameters (general case)

Parameter	Value	Variables
s_0^*	—	$\frac{A^*}{m^*}$ ω^* γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_0	—	γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_1	$\psi_1^{\text{opt}}(J)$	J $C_{L,0}$ ΔC_D
δ_0	90°	None

4 Results

4.1 Hover

4.2 Prescribed Trajectory

4.3 Prey Pursuit

References

[1] Akira Azuma, Soichi Azuma, Isao Watanabe, and Toyohiko Furuta. Flight mechanics of a dragonfly. *Journal of Experimental Biology*, 116:79–107, 1997.